Gulla Mitra

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.30
Sri Chaitanya Junior College, Vijayawada	Class XII (Board of Intermediate Education, AP)	2024	98.20%
Sri Chaitanya High School, Vijayawada	Class X (Board of Secondary Education, AP)	2022	98.17%

SCHOLASTIC ACHIEVEMENTS

- Received the LPUNEST Merit Scholarship for securing a good rank in the LPUNEST 2024 examination.

SKILLS

- **Programming Languages:** C, C++, Java, JavaScript, SQL, Assembly (RISC-V)
- **Databases:** PostgreSQL, MySQL
- **Operating Systems:** Ubuntu, Windows
- **Web Technologies:** HTML, CSS, React
- **Tools:** Git, GitHub, LaTeX, Vim, Solid Edge

PROJECTS

Process Scheduling and Virtual Memory Management in xv6

- Implemented a **System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ)** scheduler with dynamic priority adjustment and periodic **priority boosting** to prevent starvation, along with **custom system calls** for per-process system call accounting and scheduling statistics.
- Implemented an **in-memory swap mechanism** with lazy page allocation, page fault handling, **Clock-based page replacement,** and **scheduler-aware page eviction**.
- Built a **RAID-backed swap system** to replace the in-memory swap mechanism, with **FCFS and SSTF disk scheduling** for swap operations.

Arbitrary Precision Arithmetic Library

- Implemented **arbitrary-precision integer and floating-point arithmetic** using **string-based number representation** with custom algorithms for addition, subtraction, multiplication, and division supporting accurate computations with very large numbers and high-precision decimal values.
- Designed and packaged the project as a reusable **object-oriented Java library (JAR)** for integration into other Java applications.

Optimization of Matrix Multiplication using FPU Registers

- Optimized **blocked matrix multiplication** by using fmv instructions to store temporary accumulators in **FPU registers** instead of the stack, reducing repeated **ld/sd operations**.
- Compared the **stack-based** and **FPU register-based** implementations on **gem5** under different cache latencies.
- Achieved better performance at higher **cache latencies** by using FPU registers to reduce memory accesses and **memory stall cycles**.

Semaphores Synchronization

- Implemented programs to solve the **Fair Readers-Writers** Problem (i.e., Reader and Writer are given equal preference) and **Writers Preference** problem using **semaphores** and compared their performance based on **average and worst-case waiting time** to access the **critical section (**shared resources).

RELEVANT COURSES

- **Computer Science:** Introduction to Programming, Software Development Fundamentals, Operating Systems, Computer Architecture, Database Management Systems, Data Structures, Algorithms, Discrete Mathematics, Theory of Computation, Introduction to Compilers.

- **Mathematics & Other:** Calculus, Linear Algebra, Probability and Random Variables, Transform Techniques, Introduction to Statistics, Introduction to Artificial Intelligence, Digital Fabrication, Communication Skills.

EXTRACURRICULAR

- Member of the **National Service Scheme (NSS).**

- **Hobbies & Interests:** Cricket, Movies, Travelling